

MIXED DILOGARITHM PERIODS ON THE HOSTS $1/\sqrt{1 \mp 4mx}$ FROM THE CALEGARI–DIMITROV–TANG HOLONOMY BOUND

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ABSTRACT. Calegari, Dimitrov and Tang proved that a power series $f \in \mathbb{Q}[[x]]$ whose n -th coefficient has denominator dividing $[1, \dots, n][1, \dots, n/2]$, which is holomorphic on $\mathbb{C} \setminus [1, \infty)$ and continues meromorphically along all paths in $\mathbb{P}^1 \setminus \{0, \delta, 1, \infty\}$, has no monodromy at the fourth puncture δ ; they applied it with $\delta = \frac{1}{4}$ to prove that $1, \pi/\sqrt{3}$ and $3L(2, \chi_{-3}) - \frac{\pi}{\sqrt{3}} \log 3$ are linearly independent over $\mathbb{Q}$. We observe that δ is arbitrary in their theorem, and that an elementary p -adic lemma (“half-integration”: $\int h k$ has the required denominator type whenever $h \in \mathbb{Z}[[t]]$ and k has denominators $[1, \dots, j]$) supplies the needed functions on every host $H_A = 1/\sqrt{1 - 4mx}$, $m \in \mathbb{Z} \setminus \{0\}$. This gives, for every $m \geq 1$, the $\mathbb{Q}$ -linear independence of $1, \theta_m/\sqrt{4m-1}$ and $(\theta_m \log \frac{4m-1}{m} - 2\text{Cl}_2(\pi - \theta_m))/\sqrt{4m-1}$, where $\theta_m = 2 \arctan(4m-1)^{-1/2}$ and Cl_2 is Clausen’s function, i.e. the Bloch–Wigner dilogarithm at a point of $\mathbb{Q}(\sqrt{-(4m-1)})$; and, for $4m+1 = c^2$, the independence of $1, \log \frac{c+1}{c-1}$ and $2\text{Li}_2(\frac{c-1}{2c}) - \frac{\pi^2}{6} + (\text{products of logarithms})$, in particular $2\text{Li}_2(\frac{1}{3}) - \frac{\pi^2}{6} + \log^2 3 - \frac{1}{2} \log^2 2 \notin \mathbb{Q} + \mathbb{Q} \log 2$. The same argument runs on the Kummer hosts $(1 - Nx)^{-1/k}, k \text{ rad}(k) \mid N$, with scalar monodromy ζ_k , and yields for each sector $a < k$ the $\mathbb{Q}$ -linear independence of $1, k \int_0^1 v^{a-1} dv/(D+v^k)$ and $k \int_0^1 v^{a-1} \log \frac{D+v^k}{N} dv/(D+v^k)$, $D = N-1$: mixed dilogarithm periods over $\mathbb{Q}(\zeta_{2k}, D^{1/k})$. We also record that the function H_C of CDT’s Lemma on the $\delta = \frac{1}{4}$ host has denominator type $[1, \dots, n]^2$ rather than the stated $[1, \dots, n][1, \dots, n/2]$, so that the π^2 clause of their mixed theorem is not covered by their argument.

1. STATEMENT

Write $[1..n] = \text{lcm}(1, \dots, n)$ and $[1..n/2] = [1..[n/2]]$. A series $\sum a_n x^n \in \mathbb{Q}[[x]]$ has type $T(n)$ if $T(n)a_n \in \mathbb{Z}$ for all n . $\text{Cl}_2(\phi) = \sum_{n \geq 1} \sin(n\phi)/n^2$ is Clausen’s function and $D(z) = \text{Im Li}_2(z) + \arg(1-z) \log |z|$ the Bloch–Wigner dilogarithm, so that $\text{Cl}_2(\phi) = D(e^{i\phi})$.

The input is the following theorem of Calegari, Dimitrov and Tang [1, Theorem `noextrassmall`, part (*)], proved there from their basic holonomy bound with the bivalent map $\varphi(z) = 8(z + z^3)/(1+z)^4$, $\varphi'(0) = 8$, Bost–Charles integral $\log 8 + 4G/\pi$, and the five functions $1, \log(1-x), \log^2(1-x), f, f(x/(x-1))$ of denominator rate $\tau = 69/50$, for which the bound reads $5 \leq 4.640 \dots$

Theorem 1.1 (CDT). *Let $\delta \in (-\infty, 1)$. Let $f \in \mathbb{Q}[[x]]$ have type $[1..n][1..n/2]$, be holomorphic in $\mathbb{C} \setminus [1, \infty)$, and continue as a meromorphic function along all paths in $\mathbb{P}^1 \setminus \{0, \delta, 1, \infty\}$. Then f continues meromorphically along all paths in $\mathbb{P}^1 \setminus \{0, 1, \infty\}$.*

Nothing in the proof depends on the value of δ (only $\delta \neq 2$ is used, so that δ and $\delta/(\delta-1)$ are distinct). For the reader’s convenience, and because we shall need two small extensions, here is that proof reduced to its inputs. *Input 1* (CDT Theorem `basic main`): for an $m \times r$ array $\mathbf{b}$ of denominator rates with row sums σ_i (columns of the form $0 = \dots = 0 < b_j = \dots = b_j$), $\tau(\mathbf{b}) = m^{-2} \sum_i (2i-1)\sigma_i$, and a holomorphic $\varphi: (\mathbb{D}, 0) \rightarrow (\mathbb{C}, 0)$ with $|\varphi'(0)| > e^{\sigma_m}$, any $\mathbb{Q}(x)$ -linearly independent $f_1, \dots, f_m \in \mathbb{Q}[[x]]$ of types $\prod_j [1..b_{i,j}n]$ with meromorphic pullbacks $f_i \circ \varphi$ on $\mathbb{D}$

satisfy $m \leq \iint_{\mathbb{T}^2} \log |\varphi(z) - \varphi(w)| d\mu d\mu / (\log |\varphi'(0)| - \tau(\mathbf{b}))$. *Input 2:* $\varphi(z) = 8(z + z^3)/(1 + z)^4$ maps $\mathbb{D}$ onto $\mathbb{C} \setminus \{1\}$, is a bijection $(-1, 1) \rightarrow (-\infty, 1)$ and a conformal isomorphism of each open half-disc onto $\mathbb{C} \setminus (-\infty, 1]$, has $\varphi' \neq 0$ on $\mathbb{D}$, $\varphi'(0) = 8$, and Bost–Charles integral $\log 8 + 4G/\pi$ (CDT Lemma **bivalent BC integral**, via Smyth’s $m(1 + x + y - xy) = 2G/\pi$). *Input 3* (CDT Proposition **overconvergence**): if g is holomorphic on $\varphi(\Omega)$ for a contractible $\Omega \ni 0$ on which φ is univalent with $\varphi^{-1}(\Sigma^0) \subset \Omega$, and φ omits Σ^1 , then $g \circ \varphi$ converges on $\mathbb{D}$; for the bivalent map and $\Sigma^0 \subset (-\infty, 1)$ a thin neighbourhood of the diameter serves as Ω , since real $w < 1$ has a single preimage in $\mathbb{D}$. *Proof of Theorem 1.1:* if f had monodromy at δ on some sheet, then with $\iota(x) = x/(x - 1)$ (which fixes 0, swaps $1 \leftrightarrow \infty$, preserves $(-\infty, 1)$ and the type $[1..n][1..n/2]$) the five functions $1, \log(1 - x), \log^2(1 - x), f, f \circ \iota$ would be $\mathbb{C}(x)$ -independent, of types $1, [1..n], [1..n][1..n/2], [1..n][1..n/2], [1..n][1..n/2]$, so $\tau = (0 + 3 + \frac{3}{2}(5 + 7 + 9))/25 = 69/50$, with holomorphic pullbacks by Input 3 ($\Sigma^1 = \{1, \infty\}$, $\Sigma^0 = \{0, \delta, \iota\delta\}$); Input 1 gives $5 \leq 3.2452/(\log 8 - 1.38) = 4.64$. Two extensions come for free: (a) δ may be replaced by any finite $S \subset (-\infty, 1)$, f being required to be holomorphic on $\mathbb{C} \setminus [1, \infty)$ and meromorphically continuable on $\mathbb{P}^1 \setminus (\{0, 1, \infty\} \cup S)$ (take $\Sigma^0 = \{0\} \cup S \cup \iota S$); (b) the contradiction persists for f of any type of rate $\sigma \leq 1.579$, e.g. $[1..n][1..n/2][1..n/13]$.

CDT apply the theorem once, with $\delta = \frac{1}{4}$. We apply it to every host

$$(1) \quad H_A(x) = \frac{1}{\sqrt{1 - 4mx}} = \sum_{n \geq 0} \binom{2n}{n} m^n x^n \in \mathbb{Z}[[x]], \quad m \in \mathbb{Z} \setminus \{0\}, \quad \delta = \frac{1}{4m},$$

with the puncture at 1 introduced by the kernels $\frac{1}{1-t}$ and $\frac{\log(1-t)}{1-t}$:

$$H_B = H_A(x) \int_0^x \frac{H_A(t) dt}{1-t}, \quad H_D = H_A(x) \int_0^x \frac{H_A(t) \log(1-t)}{1-t} dt.$$

Theorem 1.2. Let $m \geq 1$, $D' = 4m - 1$, $\theta_m = 2 \arctan(1/\sqrt{D'})$, so that $\pi - \theta_m = 2 \arctan \sqrt{4m - 1}$. The three numbers

$$1, \quad \frac{\theta_m}{\sqrt{D'}}, \quad \frac{\theta_m \log \frac{4m-1}{m} - 2 \operatorname{Cl}_2(\pi - \theta_m)}{\sqrt{D'}}$$

are linearly independent over $\mathbb{Q}$. Equivalently, $2 \operatorname{Cl}_2(2 \arctan \sqrt{4m - 1}) - \theta_m \log \frac{4m-1}{m} \notin \mathbb{Q} \sqrt{4m - 1} + \mathbb{Q} \theta_m$.

For $m = 1$ this is the three-element part of CDT’s Theorem **mixed**: $\theta_1 = \pi/3$ and $2 \operatorname{Cl}_2(2\pi/3) = \sqrt{3} L(2, \chi_{-3})$. For $m \geq 2$ the angle θ_m is not a rational multiple of π : $e^{i\theta_m} = w_m := \frac{(\sqrt{D'} + i)^2}{4m} = \frac{2m-1+\sqrt{-D'}}{2m} \in \mathbb{Q}(\sqrt{-(4m-1)})$, and $\operatorname{Cl}_2(\pi - \theta_m) = D(-\bar{w}_m)$ is the volume of the ideal hyperbolic tetrahedron of shape $-\bar{w}_m$. The discriminants $-(4m-1)$ run through $-3, -7, -11, -15, -19, \dots$; for $m = 2, 3, 5, 11, 17, 41$ the field is a Heegner field. Only $m = 1$ produces an L -value: for $m \geq 2$ one checks $w_m \wedge (1 - w_m) \neq 0$ in $\Lambda^2 \mathbb{Q}(\sqrt{-D'})^\times \otimes \mathbb{Q}$ (for $m = 2$: $w_2 = -\bar{\pi}/\pi$ and $1 - w_2 = 1/\pi$ with $\pi = \frac{1+\sqrt{-7}}{2}$, so $w_2 \wedge (1 - w_2) = -\bar{\pi} \wedge \pi \neq 0$), so w_m is not in the Bloch group and its dilogarithm is not expected to be a rational multiple of $\sqrt{D'} L(2, \chi_{-D'})$; numerically it is not.

Theorem 1.3. Let $m \geq 1$, $c = \sqrt{4m + 1}$, $\varepsilon = \frac{c+1}{c-1}$, and

$$R_m = 2 \operatorname{Li}_2\left(\frac{c-1}{2c}\right) - \frac{\pi^2}{6} + \log(2c) \log \varepsilon + \log \frac{c+1}{2c} \log \frac{c-1}{2c} - \frac{1}{2} \log^2(c+1) + \frac{1}{2} \log^2(c-1).$$

Then $1, \frac{1}{c} \log \varepsilon, \frac{1}{c} R_m$ are linearly independent over $\mathbb{Q}$. In particular, for $4m + 1 = c^2$ a square:

$$c = 3: \quad 2 \operatorname{Li}_2\left(\frac{1}{3}\right) - \frac{\pi^2}{6} + \log^2 3 - \frac{1}{2} \log^2 2 \notin \mathbb{Q} + \mathbb{Q} \log 2,$$

$$c = 5: \quad 2 \operatorname{Li}_2\left(\frac{2}{5}\right) - \frac{\pi^2}{6} + \log 10 \log \frac{3}{2} + \log \frac{3}{5} \log \frac{2}{5} - \frac{1}{2} \log^2 6 + \frac{1}{2} \log^2 4 \notin \mathbb{Q} + \mathbb{Q} \log \frac{3}{2},$$

and similarly with $\text{Li}_2(\frac{3}{7}), \text{Li}_2(\frac{4}{9}), \dots$ for $c = 7, 9, \dots$.

The irrationality of $\text{Li}_2(1/n)$ is known for all integers n outside $\{-4, -3, -2, 2, 3, 4, 5\}$ (Hata, Rhin–Viola) and open inside; CDT [1, § “Integral holonomic modules”] single out these values. Theorem 1.3 at $c = 3$ is, to our knowledge, the first statement of this kind about $\text{Li}_2(1/3)$.

2. DENOMINATORS: THE HALF-INTEGRATION LEMMA

Lemma 2.1. *Let $h \in \mathbb{Z}[[t]]$ and $k = \sum_{j \geq 0} k_j t^j \in \mathbb{Q}[[t]]$ with $[1..j]k_j \in \mathbb{Z}$ for all j (where $[1..0] = 1$). Then $J(x) = \int_0^x h(t)k(t) dt$ has type $[1..n][1..n/2]$, and so does hJ .*

Proof. $[x^n]J = \frac{1}{n} \sum_{j=0}^{n-1} h_{n-1-j} k_j$. Fix a prime p ; put $e = \lfloor \log_p n \rfloor$, $e' = \lfloor \log_p(n/2) \rfloor$, $f = v_p(n)$, $g = \lfloor \log_p(n-1) \rfloor$. As $j \leq n-1$, $v_p(k_j) \geq -g$, so $v_p([x^n]J) \geq -(f+g)$, and we must show $f+g \leq e+e'$; note $e' \geq e-1$ always. If $n = p^e$ then $f = e$, $g = e-1 = e'$, and equality holds. If $n \neq p^e$ then $g = e$; if moreover $f = e$ then $p^e \mid n$, $n \neq p^e$ force $n \geq 2p^e$, so $e' = e$ and $f+g = 2e = e+e'$; if $f \leq e-1$ then $f+g \leq 2e-1 \leq e+e'$. Multiplication by $h \in \mathbb{Z}[[t]]$ preserves the type because $[1..n']$ divides $[1..n][1..n/2]$ for $n' \leq n$. $\square$

Corollary 2.2. *On every host (1), H_A has type 1, H_B has type $[1..n]$, and H_D has type $[1..n][1..n/2]$; more generally $H[k] := H_A \int_0^x H_A k dt$ has type $[1..n][1..n/2]$ for $k = r(t) \log(1-t)$ and type $[1..n]$ for $k = r(t)$, whenever $r \in \mathbb{Z}[t, \frac{1}{1-t}]$.*

Proof. $\frac{\log(1-t)}{1-t} = -\sum_{j \geq 1} \mathcal{H}_j t^j$ with harmonic numbers $\mathcal{H}_j$, and $[1..j]\mathcal{H}_j \in \mathbb{Z}$; multiplying k by $r \in \mathbb{Z}[t, \frac{1}{1-t}] \subset \mathbb{Z}[[t]]$ preserves the hypothesis of Lemma 2.1. For $k = r$ the coefficient of x^n in $\int_0^x H_A r$ is $\frac{1}{n}$ times an integer. $\square$

The lemma is sharp in the following sense. For $k = \frac{\log(1-t)}{t} = -\sum_{j \geq 0} \frac{t^j}{j+1}$ the hypothesis fails exactly at $j = p-1$, where the $\frac{1}{p}$ of k_{p-1} meets the $\frac{1}{p}$ of the outer $\frac{1}{n}$ at $n = p$. Accordingly CDT’s function $H_C = H_A \int_0^x H_A \log(1-t)/t dt$ on the $m = 1$ host has $H_C = -x - \frac{13}{4}x^2 - \frac{197}{18}x^3 - \dots$, its excess over $[1..n][1..n/2]$ is unbounded (it is $2 \cdot 7$ at $n = 8$ and $2 \cdot 11 \cdot 13$ at $n = 16$), and its true type is $[1..n]^2$; see Remark 4.1.

3. THE FOLD AND THE SHEETS

Fix $m \geq 1$ (the real family $m < 0$ is identical, with $\sqrt{1-4mx} = \sqrt{1+4|m|x}$; we spell out the periods in §4). On a neighbourhood of $[0, \delta]$ put $u = \sqrt{1-4mx}$, so $x = \frac{1-u^2}{4m}$, $H_A = \frac{1}{u}$, $1-x = \frac{D'+u^2}{4m}$.

Lemma 3.1 (fold decomposition). *Let $k \in \mathbb{Q}[[t]]$ extend holomorphically to $\mathbb{C} \setminus \{1\}$ and put $\kappa(v) = k(\frac{1-v^2}{4m})$, an even function holomorphic on $|v| < \sqrt{D'}$. Then*

$$H[k](x) = \frac{1}{2mu} \int_u^1 \kappa(v) dv = c[k] H_A(x) + E_k(x), \quad c[k] = \frac{1}{2m} \int_0^1 \kappa(v) dv = \int_0^\delta H_A(t)k(t) dt,$$

where $E_k = -\frac{1}{2m} \cdot \frac{1}{u} \int_0^u \kappa$ is an even holomorphic function of u on $|u| < \sqrt{D'}$, hence a holomorphic function of x on the disc $|x-\delta| < D'/(4m) = 1-\delta$, which contains 0 and has 1 on its boundary.

Proof. Substituting $t = \frac{1-v^2}{4m}$ in $\int_0^x H_A k dt$ gives $H_A(t) dt = -\frac{dv}{2m}$ and $v: 1 \rightarrow u$. Since κ is even, $\int_0^u \kappa$ is odd in u . $\square$

Hence the local monodromy T_δ at the fold satisfies $T_\delta H_A = -H_A$ and $T_\delta H[k] = H[k] - 2c[k]H_A$.

Corollary 3.2. *If $a_0 + \sum_i a_i c[k_i] = 0$ with $a_i \in \mathbb{Q}$ then $f = a_0 H_A + \sum_i a_i H[k_i] = \sum_i a_i E_{k_i}$ is holomorphic on $|x - \delta| < 1 - \delta$, hence on the simply connected slit plane $\mathbb{C} \setminus [1, \infty)$; it continues holomorphically along all paths in $\mathbb{P}^1 \setminus \{0, \delta, 1, \infty\}$; and after clearing the denominators of the a_i it has type $[1..n][1..n/2]$. So Theorem 1.1 applies to f .*

Let T_1 denote continuation along a loop based near 0 encircling 1 once and not δ , with orientation $\epsilon = \pm 1$. The loop does not change the branch of H_A , and $H_A(1)^2 = 1/(1 - 4m)$, so $H_A(1) \neq 0$.

Lemma 3.3. $T_1 H_A = H_A$; $T_1 H_B = H_B + r_B H_A$ with $r_B = -2\pi i \epsilon H_A(1) \neq 0$; and $T_1 H_D = H_D + 2\pi i \epsilon H_B + c_1 H_A$ for a constant c_1 .

Proof. Write $I_B = \int_0^x H_A/(1-t)$ and $I_D = \int_0^x H_A \log(1-t)/(1-t)$. Near $t = 1$, $H_A(t) = H_A(1) + (t-1)\tilde{H}(t)$ with $\tilde{H}$ analytic, so $I_B = -H_A(1) \log(1-x) + (\text{analytic at } 1)$ and $I_D = -\frac{1}{2} H_A(1) \log^2(1-x) + B(x) \log(1-x) + A(x)$ with A, B analytic at 1. Continuation replaces $\log(1-x)$ by $\log(1-x) + 2\pi i \epsilon$, giving $T_1 I_B = I_B + r_B$ and $T_1 I_D = I_D - 2\pi i \epsilon H_A(1) \log(1-x) + (\text{analytic}) = I_D + 2\pi i \epsilon I_B + (\text{analytic})$. Since $\{1, I_B, I_D\}$ is a basis of solutions of the third-order operator annihilating I_D (its homogeneous part annihilates $H_A/(1-t)$ and $H_A \log(1-t)/(1-t)$), and monodromy preserves this solution space with constant coefficients, the analytic remainder is a constant. Multiply by H_A . $\square$

Proposition 3.4. *Let $f = aH_A + bH_B + dH_D$ with $a, b, d \in \mathbb{Q}$ and $a + bc_B + dc_D = 0$, where $c_B = c[\frac{1}{1-t}]$, $c_D = c[\frac{\log(1-t)}{1-t}]$. Then for every $k \geq 1$*

$$(T_\delta - 1)T_1^k f = -2H_A \left[k(br_B + dc_1 + 2\pi i \epsilon dc_B) + \pi i \epsilon dr_B k(k-1) \right].$$

If $(a, b, d) \neq 0$, this is nonzero for $k = 1$ or $k = 2$.

Proof. Iterating Lemma 3.3, $T_1^k H_B = H_B + kr_B H_A$ and $T_1^k H_D = H_D + 2\pi i \epsilon k H_B + (kc_1 + 2\pi i \epsilon r_B \binom{k}{2}) H_A$; apply $T_\delta - 1$, which kills f , sends $H_A \mapsto -2H_A$ and $H_B \mapsto -2c_B H_A$. If the bracket vanishes for $k = 1, 2$ then its $k(k-1)$ -coefficient $\pi i \epsilon dr_B$ vanishes, so $d = 0$ ($r_B \neq 0$), then $br_B = 0$ so $b = 0$, then $a = 0$. $\square$

Theorem 3.5. *On every host (1) the numbers $1, c_B, c_D$ are linearly independent over $\mathbb{Q}$.*

Proof. A relation $a + bc_B + dc_D = 0$ with $(a, b, d) \in \mathbb{Q}^3 \setminus 0$ gives, by Corollary 3.2, a function f to which Theorem 1.1 applies; so f continues meromorphically along every path in $\mathbb{P}^1 \setminus \{0, 1, \infty\}$, in particular along k loops around 1 followed by a loop around δ , which must return to the same germ. Proposition 3.4 produces instead a nonzero multiple of H_A , which has a genuine branch point at δ . $\square$

4. THE PERIODS

4.1. Imaginary family, $m \geq 1$ (Theorem 1.2). With $a = \sqrt{D'}$ and $\phi = \arctan(1/a) = \theta_m/2$, Lemma 3.1 gives

$$c_B = \frac{1}{2m} \int_0^1 \frac{4m dv}{D' + v^2} = \frac{\theta_m}{a}, \quad c_D = 2 \int_0^1 \frac{\log \frac{D' + v^2}{4m}}{D' + v^2} dv = \frac{2}{a} \left[\phi \log \frac{D'}{4m} + \int_0^\phi \log(1 + \tan^2 \psi) d\psi \right].$$

Since $\int_0^\phi \log(2 \cos \psi) d\psi = \frac{1}{2} \text{Cl}_2(\pi - 2\phi)$, the last integral equals $2\phi \log 2 - \text{Cl}_2(\pi - 2\phi)$, and

$$c_D = \frac{\theta_m \log \frac{D'}{m} - 2 \text{Cl}_2(\pi - \theta_m)}{\sqrt{D'}}.$$

Theorem 1.2 is Theorem 3.5 with these values. (Both closed forms were checked to 50 digits for $m = 1, 2, 3, 5, 11$.)

4.2. Real family, $m \leq -1$ (Theorem 1.3). Write $m = -m'$ with $m' \geq 1$, $\delta = -\frac{1}{4m'} < 0$, $u = \sqrt{1 + 4m'x}$, $1 - x = \frac{c^2 - u^2}{4m'}$ with $c^2 = 4m' + 1$. Lemma 3.1 and Proposition 3.4 hold verbatim (now $H_A(1) = 1/c$ and $r_B = -2\pi i\epsilon/c \neq 0$), and

$$c_B = 2 \int_0^1 \frac{dv}{c^2 - v^2} = \frac{1}{c} \log \frac{c+1}{c-1}, \quad c_D = 2 \int_0^1 \frac{\log \frac{c^2 - v^2}{4m'}}{c^2 - v^2} dv = \frac{R_{m'}}{c},$$

using $\frac{1}{c^2 - v^2} = \frac{1}{2c} \left(\frac{1}{c-v} + \frac{1}{c+v} \right)$, $\int \frac{\log(2c-w)}{w} dw = \log(2c) \log w - \text{Li}_2\left(\frac{w}{2c}\right)$ and the reflection formula $\text{Li}_2(z) + \text{Li}_2(1-z) = \frac{\pi^2}{6} - \log z \log(1-z)$ to merge $\text{Li}_2\left(\frac{c+1}{2c}\right)$ into $\text{Li}_2\left(\frac{c-1}{2c}\right)$. (Checked to 50 digits for $m' = 1, 2, 3, 6, 12$.) Theorem 1.3 follows. For $c = 3$: $\log \varepsilon = \log 2$ and $R = 2\text{Li}_2\left(\frac{1}{3}\right) - \frac{\pi^2}{6} + \log 6 \log 2 + \log \frac{2}{3} \log \frac{1}{3} - \frac{1}{2} \log^2 4 + \frac{1}{2} \log^2 2 = 2\text{Li}_2\left(\frac{1}{3}\right) - \frac{\pi^2}{6} - \frac{1}{2} \log^2 2 + \log^2 3$.

4.3. The period space of the whole $\tau = \frac{3}{2}$ layer. Let $k = r(t) \log(1-t)$ or $k = r(t)$ with $r \in \mathbb{Q}[t, \frac{1}{1-t}]$ (Corollary 2.2). Then $c[k] = \frac{1}{2m} \int_0^1 \tilde{r}(v^2) [\log(D' + v^2) - \log 4m] dv$ with $\tilde{r}(v^2) = r\left(\frac{1-v^2}{4m}\right) \in \mathbb{Q}[v^2, \frac{1}{D'+v^2}]$. Splitting $\tilde{r}$ into a polynomial part and pole parts $(D' + v^2)^{-j}$: the polynomial part contributes $\mathbb{Q} + \mathbb{Q}\sqrt{D'}\theta_m$ (the stand-alone $\log(D' + 1) = \log 4m$ produced by $\int v^{2i} \log(D' + v^2)$ cancels against $-\log(4m) \int v^{2i}$); $j = 1$ contributes $\mathbb{Q}c_D + \mathbb{Q}c_B$; and $j \geq 2$ is handled by differentiating $G(s) = \int_0^1 \frac{\log(s+v^2)}{s+v^2} dv = \frac{1}{\sqrt{s}} [\phi \log s + 2\phi \log 2 - \text{Cl}_2(\pi - 2\phi)]$, $\phi = \arctan s^{-1/2}$, in s : as $\frac{d}{ds} \text{Cl}_2(\pi - 2\phi) = 2\phi' \log(2 \cos \phi)$ and $\log s - 2 \log \cos \phi = \log(s+1)$, the derivative reproduces the same bracket up to rational factors plus a stand-alone $\log(s+1) = \log 4m$ that again cancels. Hence every period of the layer lies in

$$V_m = \mathbb{Q} \oplus \mathbb{Q} \frac{\theta_m}{\sqrt{D'}} \oplus \mathbb{Q} \sqrt{D'} \theta_m \oplus \mathbb{Q} \frac{Q_m}{\sqrt{D'}}, \quad Q_m = 2\text{Cl}_2(\pi - \theta_m) - \theta_m \log \frac{D'}{m},$$

($k = \log(1-t)$ realises $c = \frac{\sqrt{D'}\theta_m - 2}{2m}$), and since $\mathbb{Q} \frac{\theta_m}{\sqrt{D'}} + \mathbb{Q} \sqrt{D'} \theta_m = \theta_m \mathbb{Q}(\sqrt{D'})$, Theorem 1.2 is the strongest statement the $\tau = \frac{3}{2}$ layer can give: the pure Clausen value cannot be separated from $\theta_m \log \frac{D'}{m}$ at this level. Kernels containing H_A^{2s} , $s \geq 1$, have divergent fold integrals; their $H[k]$ acquires a pole at δ after subtracting the finite-part multiple of H_A , $(1 - 4mx)^s H[k]$ is again admissible, and the periods stay in V_m .

Remark 4.1 (on CDT's Lemma “around $1/4$ ” and Theorem “mixed”). CDT (Lemma 2.11.13 and Theorem 2.11.17 in the arXiv v2 numbering; the type condition is (2.5.3) in their Theorem 2.8.4, with no A^{n+1} slack) define $H_C = \frac{1}{\sqrt{1-4x}} \int_0^x \frac{\log(1-t)}{t\sqrt{1-4t}} dt$ and state that H_C and H_D have type $[1..n][1..n/2]$. For H_D this is Corollary 2.2; for H_C it is false (its coefficient of x^2 is $-\frac{13}{4}$, and its excess over $[1..n][1..n/2]$ grows without bound, while its type $[1..n]^2$ is exact), see §2. Their Theorem “mixed” asserts the $\mathbb{Q}$ -linear independence of $1, \pi/\sqrt{3}, \pi^2, 3L(2, \chi_{-3}) - \frac{\pi}{\sqrt{3}} \log 3$; the argument via Theorem 1.1 covers the three-element statement without π^2 (Theorem 1.2 at $m = 1$, where $c_B = \frac{\pi}{3\sqrt{3}}$, $c_D = \frac{\pi}{3\sqrt{3}} \log 3 - L(2, \chi_{-3})$), and by §4.3 no $[1..n][1..n/2]$ -function of the layer has fold period π^2 ; the value $\int_0^{1/4} \frac{\log(1-t)}{t\sqrt{1-4t}} dt = -\frac{\pi^2}{18}$ is correct but belongs to a $\tau = 2$ function. The π^2 clause therefore needs a different input.

Remark 4.2 (why these are the hosts). The mechanism of §3 needs a function with scalar monodromy -1 at the fold and integral coefficients, with the second singular point at 1 after a Möbius change of variable that preserves integrality. On the modular hosts of the sporadic Apéry rows the fold is either a cusp (second-order rows), where the monodromy of the modular form F is unipotent and $T(F \int Fk) - F \int Fk$ involves the two new functions $y_1 \int_\delta^x Fk$ and $F \int_\delta^x y_1 k$, so that no rational combination of F and finitely many $F \int Fk_i$ is fold-regular; or an

order-two elliptic point (third-order rows), where a Fricke-odd F does have monodromy -1 but the second singular point is either the Galois conjugate of the fold $((\sqrt{2} \pm 1)^4$ for Apéry's $\zeta(3)$ row) or sits at $\frac{1}{4}$ against a fold at $\frac{1}{16}$ (Domb), and moving it to 1 costs 4^n in the denominators. Lemma 2.1 itself holds for any $h \in \mathbb{Z}[[t]]$. The hosts $1/\sqrt{(1-x)(1-c^2x)}$, integral for odd c , have sign monodromy at both $\frac{1}{c^2}$ and 1 and reproduce the real family.

Corollary 4.3 (Mahler measures). *Let $P_m = m(1+x)^2 - (4m-1)y^2 \in \mathbb{Z}[x, y]$ and $m(P)$ the logarithmic Mahler measure. Then*

$$2\text{Cl}_2(\pi - \theta_m) - \theta_m \log \frac{4m-1}{m} = \pi \left[m(P_m) - \log(4m-1) \right],$$

so Theorem 1.2 says that 1 , $\theta_m/\sqrt{4m-1}$ and $\pi(m(P_m) - \log(4m-1))/\sqrt{4m-1}$ are linearly independent over $\mathbb{Q}$.

Proof. The Cassaigne–Maillot formula gives $\pi m(a+bx+cy) = \alpha \log a + \beta \log b + \gamma \log c + D(\frac{b}{a}e^{i\gamma})$ for a triangle with sides a, b, c and opposite angles α, β, γ . The isosceles triangle $(1, 1, \sqrt{(4m-1)/m})$ has apex angle γ with $\cos \gamma = 1 - \frac{4m-1}{2m} = -\frac{2m-1}{2m} = -\cos \theta_m$, i.e. $\gamma = \pi - \theta_m$, and base angles $\theta_m/2$; hence $\pi m(1+x+\sqrt{(4m-1)/m}y) = \frac{\pi-\theta_m}{2} \log \frac{4m-1}{m} + \text{Cl}_2(\pi - \theta_m)$. Now $2m(1+x+ty) = m((1+x)^2 - t^2y^2)$ (substitute $y \mapsto -y$ and multiply) and $m(\lambda P) = \log \lambda + m(P)$. (Checked to 10^{-121} for $m = 1, 2, 3, 5, 7, 11, 25$.) For $m = 1$ this is Smyth's $\pi m(1+x+y) = \text{Cl}_2(\pi/3)$ and CDT's $m((1+x+y)^4/3) = 4m(1+x+y) - \log 3$. $\square$

For $m \geq 2$ the apex angle is not a rational multiple of π and no integer-sided triangle realises it (none among the 31 079 triples $a \leq b \leq c \leq 60$); the algebraic triple is the natural one.

5. FROM SQUARE ROOTS TO k -TH ROOTS

Sections 2–3 used only that H_A has rank one with scalar monodromy at the fold. The same holds for

$$H(x) = (1 - k^k m x)^{-1/k}, \quad k \geq 2, \quad m \in \mathbb{Z} \setminus \{0\}, \quad \delta = \frac{1}{k^k m}, \quad D := k^k m - 1,$$

with monodromy ζ_k^{-1} at δ .

Lemma 5.1. $(1 - Nx)^{-j/k} \in \mathbb{Z}[[x]]$ for $N = k^k m$, $m \in \mathbb{Z}$, $j \geq 0$.

Proof. $[x^n](1 - Nx)^{-j/k} = \frac{(j/k)_n}{n!} N^n = \frac{\prod_{i \leq n} (j+ki)}{n!} k^{(k-1)n} m^n$. For $p \nmid k$ the progression $j, j+k, \dots, j+(n-1)k$ has difference prime to p , so any p^e consecutive terms contain a multiple of p^e and $v_p(\prod) \geq \sum_e \lfloor n/p^e \rfloor = v_p(n!)$. For $p \mid k$, $v_p(n!) < \frac{n}{p-1} \leq n \leq (k-1)n v_p(k)$. $\square$

For a sector $a \in \{1, \dots, k-1\}$ and a kernel κ put $H^{(a)}[\kappa] = H^a \int_0^x H^{k-a} \kappa dt$ and $c^{(a)}[\kappa] = \int_0^\delta H^{k-a} \kappa dt$. By Lemma 2.1 with $h = H^{k-a}$ and outer factor H^a (both in $\mathbb{Z}[[t]]$), $H^{(a)}[\frac{1}{1-t}]$ has type $[1..n]$ and $H^{(a)}[\frac{\log(1-t)}{1-t}]$ type $[1..n][1..n/2]$ (verified exactly for $k \leq 6$, $n \leq 150$). With $u = (1 - k^k m x)^{1/k}$, $dt = -u^{k-1} du / (k^{k-1} m)$, $H^{k-a} = u^{-(k-a)}$ and $\tilde{\kappa}(v) = \kappa(\frac{1-v^k}{k^k m})$, a function of v^k holomorphic for $|v| < D^{1/k}$,

$$H^{(a)}[\kappa] = \frac{u^{-a}}{k^{k-1} m} \int_u^1 v^{a-1} \tilde{\kappa}(v) dv = c^{(a)}[\kappa] H^a - \frac{u^{-a}}{k^{k-1} m} \int_0^u v^{a-1} \tilde{\kappa}(v) dv,$$

and the last term is a holomorphic function of u^k , hence of x , on $|x - \delta| < 1 - \delta$. So $T_\delta H^{(a)}[\kappa] = H^{(a)}[\kappa] + (\zeta_k^{-a} - 1) c^{(a)}[\kappa] H^a$, Corollary 3.2 holds with H_A replaced by H^a , and Lemma 3.3 and Proposition 3.4 go through verbatim (H is analytic at 1 with $H^{k-a}(1) = (-D)^{-(k-a)/k} \neq 0$, and $\zeta_k^{-a} - 1 \neq 0$). Since $1 - t = (D + v^k)/(k^k m)$:

Theorem 5.2. *For every $k \geq 2$, $m \in \mathbb{Z} \setminus \{0\}$ and $a \in \{1, \dots, k-1\}$, the numbers*

$$1, \quad k \int_0^1 \frac{v^{a-1} dv}{k^k m - 1 + v^k}, \quad k \int_0^1 \frac{v^{a-1} \log \frac{k^k m - 1 + v^k}{k^k m}}{k^k m - 1 + v^k} dv$$

are linearly independent over $\mathbb{Q}$.

The proof of Lemma 5.1 only used $v_p(N/k) \geq 1$ for $p \mid k$, so Theorem 5.2 holds verbatim with $k^k m$ replaced by any $N \in k \operatorname{rad}(k) \mathbb{Z} \setminus \{0\}$ and $D = N - 1$ (hosts $(1 - 9mx)^{-1/3}$, $(1 - 8mx)^{-1/4}$, $(1 - 25mx)^{-1/5}$, $\dots$). Since the power sums $\sum_j \alpha_j^i$ of the roots of $v^k + D$ vanish for $1 \leq i \leq k-1$, partial fractions give the closed form (principal branches, verified to 111 digits)

$$k \int_0^1 \frac{v^{a-1} dv}{D + v^k} = -\frac{1}{D} \sum_{j=0}^{k-1} \alpha_j^a \log \left(1 - \frac{1}{\alpha_j} \right),$$

and the period space of each sector is again exactly $\mathbb{Q} + \mathbb{Q}c_B^{(a)} + \mathbb{Q}c_D^{(a)}$ (with elementary identities such as $c^{(a)}[\log(1-t)] = \frac{kD}{Na}c_B^{(a)} - \frac{k^2}{Na^2}$), while different sectors of one host are arithmetically independent (PSLQ, 140 digits). When $D = n^k$ is a perfect k -th power the roots are n times roots of unity and the dilogarithm points lie in $\mathbb{Q}(\zeta_{2k})$; the fold-type points $\zeta_l/(\zeta_l - \zeta_j)$ are Bloch-group elements, but they depend only on ζ_j/ζ_l and their aggregate coefficient is proportional to $\sum_l \zeta_l^a = 0$, so they cancel identically. Thus *no* Catalan constant appears on the $k = 4$ hosts $(1 + 80x)^{-1/4}$, $(1 + 624x)^{-1/4}$ (its coefficient is exactly 0, checked to 320 digits); the only collapse host in our list retaining an L -value is $(1 - 9x)^{-1/3}$, where the Bloch–Wigner part of $c_D^{(1)}$ is $-\frac{15}{32}L(2, \chi_{-3})$. In every collapse host the surviving Bloch–Wigner points are the Cassaigne–Maillot points of the triangles with vertices $1, n\zeta_l, n\zeta_j$, so their content is Mahler-measure content, while an irreducible ReLi_2 part remains.

For $k = 2$, $a = 1$ this is Theorems 1.2 ($m > 0$) and 1.3 ($m < 0$). By partial fractions over the roots $\alpha_j = D^{1/k} e^{i\pi(2j+1)/k}$ of $v^k = -D$, the second number is a $\mathbb{Q}(\zeta_{2k}, D^{1/k})$ -linear combination of π and logarithms of elements of that field, and the third a combination of dilogarithms at the points $\alpha_j/(\alpha_j - \alpha_l)$, $(\alpha_j - 1)/(\alpha_j - \alpha_l)$ and products of such logarithms; for $k = 3$, $m = 1$ the field is $\mathbb{Q}(\omega, \sqrt[3]{26})$. The $k-1$ sectors of one host give $k-1$ separate statements (a relation mixing sectors is not fold-regular, the fold monodromies being multiples of the independent functions H^a).

Lemma 5.3 (classification of rank-one hosts). *Let $f \in \mathbb{Z}[[x]]$ be algebraic with a single finite singular point $\delta \neq 0$ and scalar local monodromy of exact order $k \geq 2$ there. Then $\delta = 1/N$ with $N \in \mathbb{Z}$, $k \operatorname{rad}(k) \mid N$, and $f = R(x)(1 - Nx)^{j/k}$ with $\gcd(j, k) = 1$ and $R \in \mathbb{Q}(x)$ having poles only at δ .*

Proof. f^k is single-valued and algebraic on $\mathbb{P}^1 \setminus \{\delta, \infty\}$, hence a rational function with zeros and poles only at δ, ∞ , so $f = cR_0(x)(1 - x/\delta)^{j/k}$. If $\delta = p/q$ in lowest terms, the coefficient of x^n in $(1 - x/\delta)^{j/k}$ is $\binom{j/k}{n}(-q/p)^n$, whose ℓ -adic valuation for $\ell \mid p$ is $-n v_\ell(p) + O(\log n)$, which no fixed rational factor can compensate; so $p = 1$. For a prime $p \mid k$ (hence $p \nmid j$), $[x^n](1 - Nx)^{j/k} = \pm \prod_{i < n} (j - ki)(N/k)^n/n!$ has p -adic valuation $n(v_p(N) - v_p(k)) - v_p(n!)$, negative at $n = p^e$ unless $v_p(N) \geq v_p(k) + 1$. The converse is Lemma 5.1 (whose proof only used $v_p(N/k) \geq 1$). $\square$

So the hosts of this paper are all the rank-one hosts: the $\tau = \frac{3}{2}$ theory is complete.

6. EFFECTIVE MEASURES

CDT's quantitative refinement of the bound [2, Proposition **theomainreprise**] reads: if η is realised as an Apéry limit by pairs (A_i, B_i) of series of the given type, $g_i = B_i - \eta A_i$ ($i \leq m$) are independent with meromorphic pullbacks, γm of them unconditional ($A_i = 0$), and $\varphi^* A_i, \varphi^* B_i$ converge on $|z| < \rho$, then, with $\mu = \mu_{\text{eff}}(\eta)$,

$$m \leq \frac{\text{BC}(\varphi)}{\log |\varphi'(0)| - \tau(\mathbf{b}) - (1 - \gamma) \left(\frac{2}{\mu} - \frac{1 - \gamma}{\mu^2} \right) \log \frac{1}{\rho}}$$

whenever the denominator is positive. On a host (1) (or a Kummer host, any sector) take $(A, B) = (H_A, H_D)$, so $\eta = c_D$, and the tuple $\{1, \log(1 - x), \log^2(1 - x), g, g \circ \iota\}$: $m = 5$, $\gamma = \frac{3}{5}$ (the unconditional part is exactly $\{1, \log, \log^2\}$ by CDT's Theorem 2.11.11), $\tau = \frac{69}{50}$, $\text{BC} = \log 8 + 4G/\pi$, and ρ defined by $8\rho(1 + \rho^2)/(1 - \rho)^4 = 1/N$ (the convergence radius of H_A, H_B, H_D). Solving for μ gives $\mu_* = (1 - \gamma)/(1 - \sqrt{1 - \text{gap}/L})$ with $\text{gap} = \log 8 - \tau - \text{BC}/m$, $L = \log(1/\rho)$; a larger true dimension only lowers the threshold. The pairs (H_A, H_B) ($\eta = c_B$; tuple $\{1, \log(1 - x), g, g \circ \iota\}$, $m = 4$, $\gamma = \frac{1}{2}$, $\tau = \frac{15}{16}$) and (H_D, H_B) ($\eta = c_B/c_D$) are treated identically.

Theorem 6.1. $\mu_{\text{eff}}(c_D) \leq \kappa_D(N)$, $\mu_{\text{eff}}(c_B/c_D) \leq \kappa_D(N)$ and $\mu_{\text{eff}}(c_B) \leq \kappa_B(N)$, where

host	N	ρ	$\log(1/\rho)$	κ_D	κ_B
$k = 2, m = 1$	4	0.02789	3.580	56.73	10.57
$k = 2, m = 2$	8	0.01472	4.218	66.89	12.51
$k = 2, m = 3$	12	0.01001	4.605	73.03	13.68
$k = 2, m = 5$	20	0.00610	5.100	80.90	15.18
$k = 2, m = 11$	44	0.00281	5.875	93.23	17.52
$k = 3, N = 9$	9	0.01317	4.330	68.66	12.85
$k = 4, N = 80$	80	0.00155	6.468	102.7	19.31

In particular $\mu_{\text{eff}}(L(2, \chi_{-3}) - \frac{\pi}{3\sqrt{3}} \log 3) \leq 56.73$ and $\mu_{\text{eff}}(\frac{2}{\sqrt{7}} \arctan \frac{1}{\sqrt{7}}) \leq 12.51$.

CDT's measure for the pure $L(2, \chi_{-3})$ is 24781; for the mixed number and for the periods with $m \geq 2$ no explicit measure seems to have been recorded. For $m = 1$ the elementary number is $\pi/(3\sqrt{3})$, where Salikhov's $\mu(\pi/\sqrt{3}) \leq 4.6016$ is much better. The real family and all Kummer sectors share these values, which depend only on N .

7. TWO FOLDS: ELLIPTIC HOSTS

Extension (a) of Theorem 1.1 allows hosts $H = 1/\sqrt{P(x)} \in \mathbb{Z}[[x]]$ with several real folds $\delta_1, \dots, \delta_s \in (0, 1)$; for $\deg P = 3$ with all roots in $(0, 1]$ the curve $y^2 = P$ is elliptic. Lemma 2.1 is unchanged, and near each fold H is odd in the local uniformiser $s = \sqrt{t - \delta_i}$, so $H[k] = c_i[k]H + (\text{even in } s)$ and $T_{\delta_i} H[k] = H[k] - 2c_i[k]H$ as before; the loop around 1 acts as in Lemma 3.3 with r_B replaced by $-2 \text{FP} \int_0^1 \frac{H dt}{1-t}$ (a Hadamard finite part when 1 is a branch point), which is nonzero on every host examined. Hence, for kernels $k_1, \dots, k_r$, the $s \times (r+1)$ matrix $(1 \mid c_i[k_j])$ has no nonzero rational vector in its kernel.

For $s = 2$ this is a statement about the 2×2 minors, i.e. about the *difference periods* $\int_{\delta_1}^{\delta_2} k dt/y$. In genus 0 (two branch points, P quadratic) these are elementary — for $P = (1 - 4m_1x)(1 - 4m_2x)$ the substitution $t = \frac{\delta_1 + \delta_2}{2} - \frac{\delta_2 - \delta_1}{2} \cos \phi$ gives $\Delta_B = \pi/\sqrt{D_1 D_2}$ and $\Delta_D = \Delta_B \log \frac{2s^2}{a+s}$ with $s = \sqrt{D_1 D_2}/(4\sqrt{m_1 m_2})$, $a = 1 - \frac{m_1 + m_2}{8m_1 m_2}$ — so the two-fold statement is empty there. In genus 1, for the hosts $P = (1 - x)(1 - ax)(1 - bx)$ with $\{a, b\} = \{9, 4m\}$ or $\{25, 4m\}$ (conductors

30, 42, 198, 210, $\dots$), where 1 is a Weierstrass point, the difference periods are identified numerically to 10^{-90} : with $\Delta = (a-1)(b-1)$, $f = ab/\Delta$, $c = 1/f$, and $\Omega = \int_{r_1}^{r_2} dx/y$, $\eta = \int_{r_1}^{r_2} x dx/y$,

$$\Delta_B = f(\Omega - \eta), \quad \Delta_L := \int_{r_1}^{r_2} \frac{\log(1-x) dx}{y} = \frac{1}{2}\Omega \log c, \quad \Delta_D = \frac{(a+b)\Omega - 2ab\eta}{\Delta} + \frac{f}{2}(\Omega - \eta) \log c.$$

The mechanism is translation by the 2-torsion point $(1, 0)$, an involution σ exchanging the two folds with $(1-x)(1-\sigma x) = c$; $dx/((1-x)y)$ is of the second kind (double pole without residue at the Weierstrass point). The near-fold periods $\int_0^{r_1} \log(1-x) dx/((1-x)y)$ are new transcendental, not in the span of $1, \Omega_{01}, \eta_{01}, \Omega, \eta, \Omega \log c, \eta \log c$. The two-fold theorem then reduces to a dichotomy: either $\log \frac{(a-1)(b-1)}{ab} \neq \frac{p\Omega + 4\eta}{\Omega - \eta}$ for every $p \in \mathbb{Q}$, or the near-fold combination $A_D - \left(\frac{p_0}{2} + \frac{a+b}{ab}\right)A_B$ is irrational for the unique real p_0 (which is not a rational of height $\leq 10^{12}$ on any of the ten hosts). One cannot force the far-fold condition by a rational choice of kernels: among $(1-t)^{-j}$, $j \leq 3$, the only combination with vanishing far-fold difference is the exact differential. So in genus one the method produces structure (the identities above) but no unconditional statement about the new elliptic mixed periods; the host $(1-4x)(1-8x)(1-12x)$ (conductor 24, isogenous to Boyd's $k = 2, 8$ curves), where 1 is not a branch point, has no linear identifications at all, only the Riemann bilinear relation $B_B\Omega_{23} - C_B\Omega_{12} = -2\pi u/\sqrt{231}$ with u the elliptic logarithm of the point over $x = 1$.

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